Maciej Tkacz

Junior Robotics / Simulation Engineer

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PROFESSIONAL PROFILE

Mechanical Engineering student at Cracow University of Technology and certified Software Technician combining CAD/mechanical design with Python/C++ programming, robotic simulation and hands-on prototyping. Commercial experience creating robotics simulations and digital twins in Visual Components for automotive production environments. Built an end-to-end SCARA robotic arm prototype covering CAD, FDM printing, actuator/driver selection, electronics, control software and hardware-software integration. Interested in automation, mobile/industrial robotics, drones and simulation-driven development.

KEY SKILLS

- Robotics simulation & digital twins: Visual Components, robot workcells, robot paths, process logic, collision checks, cycle flows and layout optimization.
- Programming: Python, C++, Arduino/C, JavaScript basics; automation scripts, calculation scripts, hardware-control experiments and technical troubleshooting.
- Robotics & control: kinematics fundamentals, motion basics, sensing concepts, control systems, dynamic systems modelling, automation, mechatronics and analytical mechanics coursework; industrial robot programming (Kawasaki, Epson - certified courses).
- Simulation exposure: MATLAB at university; ROS self-study in personal robotics projects; strong willingness to learn Gazebo, Isaac Sim and similar tools.
- CAD & prototyping: SolidWorks, Autodesk Inventor, Fusion 360, AutoCAD; rapid prototyping with FDM, SLA and MJF; design for additive manufacturing and reverse engineering.
- Hardware-software integration: NEMA stepper motors, TMC/DRV motor drivers, Arduino/ESP32-class microcontroller and single-board platforms, wiring, testing and iterative debugging of electromechanical prototypes.
- Teamwork: technical documentation, communication with engineering teams and collaborative problem solving in interdisciplinary environments.

SELECTED ROBOTICS PROJECT

SCARA Robotic Arm Prototype | Personal R&D Project

- Designed and built an end-to-end SCARA robotic arm prototype, covering mechanical concept, CAD modelling, kinematic layout, selection of NEMA stepper actuators and TMC/DRV drivers, microcontroller electronics and wiring.
- Optimized components for 3D printing, including PET-G and carbon-fibre reinforced materials, with focus on stiffness, assembly constraints and rapid iteration.
- Developed Python/C++ control software for motion and hardware-interaction experiments, applying forward/inverse kinematics and control-system fundamentals.
- Integrated mechanical, electronic and software subsystems through iterative assembly, testing and debugging, gaining practical hardware-software integration experience.

PROFESSIONAL EXPERIENCE

Application Engineer | AIAutomation | Jan 2025 - May 2026

- Created robotics simulations and digital twins of production workcells in Visual Components for automotive clients.
- Developed Python scripts for process optimization and for improving simulation behaviour and performance.
- Developed robot logic, motion sequences, robot paths, collision checks, cycle flows and virtual process validation.
- Optimized workcell layouts and 3D geometry for simulation needs, using SolidWorks and Autodesk Inventor.
- Analyzed customer documentation, engineering standards and technical specifications to support solution concepts aligned with production requirements.
- Collaborated with engineers and stakeholders, communicating simulation assumptions, constraints and improvement proposals.

3D Printing and CAD Design Specialist | Cubic Inch Additive Manufacturing, Piaseczno | Jun 2023 - Aug 2023

- Operated and serviced FDM, MJF and SLA 3D printers, supervising process parameters, post-processing and quality control.
- Designed and optimized CAD models in Fusion 360 and Autodesk Inventor for additive manufacturing and rapid prototyping.
- Supported implementation testing for a new SLA technology, documenting progress and technical observations.
- Coordinated production tasks in a 10-person project team, balancing quality, manufacturability and delivery constraints.

Robotics and 3D Printing Intern | ASTOR Robotics Center, Krakow | May 2022

- Assembled mechanical equipment and 3D-printed components for Kawasaki robots and Astorino educational robot platforms.
- Programmed Kawasaki industrial robots using teach pendant workflows and created basic robot motion programs.
- Tested robot and workstation operation, gaining practical exposure to robot setup, safety and hardware integration.

Web Application Development Intern | Souczek Design Studio Reklamy i Druku, Kielce | Jul 2021

- Solved technical problems in JavaScript, including implementation of an interactive order form.
- Tested and deployed web functionality in a team environment according to customer guidelines.

EDUCATION

- Cracow University of Technology - Mechanical Engineering, Engineer's degree in progress (Oct 2024 - present). Relevant coursework: dynamic systems modelling, automation/control, analytical mechanics and mechatronics.

- PKMechPower Student Research Group, Mechanical Section - CAD and 3D printing tasks, including driver seat and brake-system components; mass and strength-oriented design optimization.
- Zespol Szkol Informatycznych im. gen. Jozefa Hauke-Bosaka, Kielce - Software Technician (Sep 2019 - Apr 2024).

CERTIFICATES

- Kawasaki robot operation and programming - integrator course with certificate, ASTOR Robotics Center (May 2022).
- Epson industrial robot programming - certified course.
- Python programming courses and self-directed Python/C++ development in robotics and automation projects.